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GATE ECE · ELECTROMAGNETICS · CHAPTER-WISE PYQS

Antennas

GATE ECE Electromagnetics · Previous Year Questions with Answer Key

55	31	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	18 questions
Marks 2	26 questions
Marks 4	1 question
Marks 5	7 questions
Marks 8	1 question
Marks 10	2 questions
Answer key	at the end

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Marks 1 18 questions

Q1. GATE ECE 2026

Consider the Friis' transmission equation $P_R = \left(P_T G_T G_R \lambda^2\right)/(4\pi D)^2$, where P_R and P_T are the received and the transmitted powers, respectively.

G_T and G_R are the gain of transmitting and receiving antennas, respectively, D is the distance between the transmitting and receiving antennas, and λ is the wavelength in free space.

Given: $G_T = G_R = 1.0, \lambda = 0.30\ \mathrm{m}$ and $P_T = +10\mathrm{dBm}$.

Choose the distance (D), in km , from the following options at which the received power, $P_R = -90\mathrm{dBm}$?

(A) $\frac{15}{4\pi}$ (B) $\frac{15}{2\pi}$ (C) $\frac{75}{2\pi}$ (D) $\frac{3}{4\pi}$

Open solution

Q2. GATE ECE 2021

An antenna with a directive gain of 6 dB is radiating a total power of 16 kw . The amplitude of the electric field in free space at a distance of 8 km from the antenna in the direction of 6 dB gain(rounded off to three decimal places is____ $\mathrm{V/m}$.

Open solution

Q3. GATE ECE 2015 Set 3

The directivity of an antenna array can be increased by adding more antenna elements, as a larger number of elements

(A) Improves the radiation efficiency

(B) Increases the effective area of the antenna

(C) Results in a better impedance matching

(D) Allows more power to be transmitted by the antenna

Open solution

Q4. GATE ECE 2014 Set 4

For an antenna radiating in free space, the electric field at a distance of 1 km is found to be 12 m V/m. Given that intrinsic impedance of the free space is $120\pi\Omega$, the magnitude of average power density due to this antenna at a distance of 2 km from the antenna (in n W/m^2) is

Open solution

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Q5. GATE ECE 2014 Set 4

Open solution

Match column A with column B.

Column

1. Point electromagnetic source
2. Dish antenna
3. Yagi-Uda antenna

Column

P. Highly directional
Q. End fire
R. Isotropic

(A) 1 = P, 2 = Q, 3 = R
(B) 1 = R, 2 = P, 3 = Q
(C) 1 = Q, 2 = P, 3 = R
(D) 1 = R, 2 = Q, 3 = P

Q6. GATE ECE 2012

Open solution

The radiation pattern of an antenna in spherical co-ordinates is given by

$$F(\theta) = \cos^4\theta \ ; \ 0 \leq \theta \leq \pi/2$$

The directivity of the antenna is

(A) 10 dB
(B) 12.6 dB
(C) 11.5 dB
(D) 18 dB

Q7. GATE ECE 2008

Open solution

For a Hertz dipole antenna, the half power beam width (HPBW) in the E-plane is

(A) 360°
(B) 180°
(C) 90°
(D) 45°

Q8. GATE ECE 2006

Open solution

A transmission line is feeding 1 Watt of power to a horn antenna having gain of 10 dB. The antenna is matched to the transmission line. The total power radiated by the horn antenna into the free-space is

(A) 10 Watts
(B) 1 Watt
(C) 0.1 Watt
(D) 0.01 Watt

Q9. GATE ECE 2005

Open solution

Refractive index of glass is 1.5. Find the wavelength of a beam of light with a frequency of 10^{14} Hz in glass. Assume velocity of light is 3×10^8 m/s in vacuum.

(A) 3 μm
(B) 3mm
(C) 2 μm
(D) 1 μm

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Q10. GATE ECE 2004

Consider a lossless antenna with a directive gain of + 6 dB. If 1 m W of power is fed to it the total power radiated by the antenna will be

(A) 4 m W (B) 1 m W (C) 7 m W (D) 1/4 m W

Open solution

Q11. GATE ECE 2002

The line-of-sight communication requires the transmit and receive antennas to face each other. If the transmit antenna is vertically polarized for best reception the receiver antenna should be

(A) horizontally polarized

(B) vertically polarized

(C) at 45° with respect to horizontal polarization

(D) at 45° with respect to vertical polarization

Open solution

Q12. GATE ECE 2000

If the diameter of a $\lambda/2$ dipole antenna is increased from $\lambda/100$ to $\lambda/50$ then is

(A) bandwidth increases (B) bandwidth decreases

(C) gain increases (D) gain decreases

Open solution

Q13. GATE ECE 1998

The radiation resistance of a circular loop of one turn is 0.01 Ω. The radiation resistance of five turns of such a loop will be

(A) 0.002 Ω (B) 0.01 Ω (C) 0.05 Ω (D) 0.25 Ω

Open solution

Q14. GATE ECE 1998

An antenna in free space receives 2μ W of power when the incident electric field is 20 m V/m rms. The effective aperture of the antenna is

(A) 0.005 m^2 (B) 0.05 m^2 (C) 1.885 m^2 (D) 3.77 m^2

Open solution

Q15. GATE ECE 1998

The far field of an antenna varies with distance r as

(A) 1/r (B) 1 / r^2 (C) 1 / r^3 (D) $\frac{1}{\sqrt{r}}$

Open solution

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Q16. GATE ECE 1998

The vector $\vec{H}$ in the far field of an antenna satisfiels

(A) $\nabla . \vec{H} = 0 \; and \; \nabla \times \vec{H} = 0$

(B) $\nabla . \vec{H} \neq 0 \; and \; \nabla \times \vec{H} \neq 0$

(C) $\nabla . \vec{H} = 0 \; and \; \nabla \times \vec{H} \neq 0$

(D) $\nabla . \vec{H} \neq 0 \; and \; \nabla \times \vec{H} = 0$

Open solution

Q17. GATE ECE 1995

An antenna when radiating, has a highly directional radiation pattern. When the antenna is receiving its radiation pattern

(A) is more directive

(B) is less directive

(C) is the same

(D) exhibits no directivity at all

Open solution

Q18. GATE ECE 1994

For a dipole antenna

(A) the radiation intensity is maximum along the normal to the dipole axis

(B) the current distribution along its length is uniform irrespective of the length

(C) the effective length equals its physical length

(D) the input impedance is independent of the location of the feed-point

Open solution

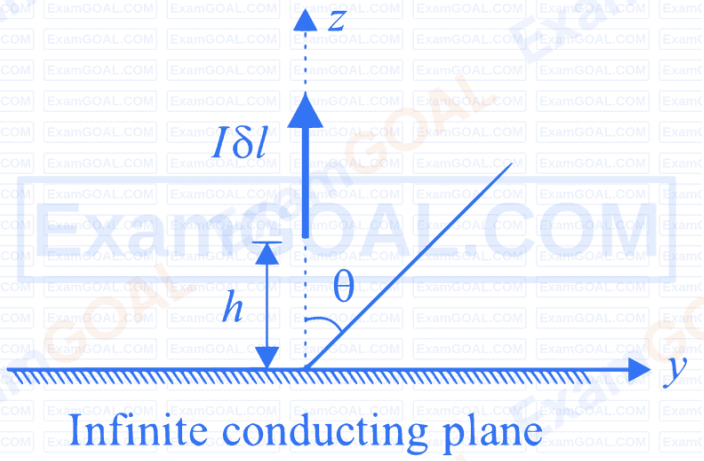
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Marks 2 26 questions

Q19. GATE ECE 2020

Open solution

For an infinitesimally small dipole in free space, the electric field E_θ in the far field is proportional to $\frac{e^{-jkr}}{r}\sin\theta$, where $k = \frac{2\pi}{\lambda}$. A vertical infinitesimally small electric dipole ($\delta l \ll \lambda$) is placed at a distance $h(h > 0)$ above an infinite ideal conducting plane, as shown in the figure. The minimum value of h, for which one of the maxima in the far field radiation pattern occurs at $\theta = 60^\circ$, is



(A) 0.75λ (B) 0.25λ (C) 0.5λ (D) λ

Q20. GATE ECE 2017 Set 1

Open solution

A half wavelength dipole is kept in the x-y plane and oriented along 45° from the x-axis. Determine the direction of null in the radiation pattern for $0 \le \phi \le \pi$. Here the angle $\theta\,(0 \le \theta \le \pi)$ is measured from the z-axis, and the angle $\phi\,(0 \le \phi \le 2\pi)$ is measured from the x-axis in the x-y plane.

(A) $\theta = 90^\circ,\ \phi = 45^\circ$ (B) $\theta = 45^\circ,\ \phi = 90^\circ$

(C) $\theta = 90^\circ,\ \phi = 135^\circ$ (D) $\theta = 45^\circ,\ \phi = 135^\circ$

Q21. GATE ECE 2016 Set 1

Open solution

An antenna pointing in a certain direction has a noise temperature of $50K$. The ambient temperature is $290K$. The antenna is connected to a pre-amplifier that has a noise figure of 2 dB and an available gain of 40 dB over an effective bandwidth of $12\ MHz$. The effective input noise temperature T_e for the amplifier and the noise power P_{ao} at the output of the preamplifier, respectively, are

(A) $T_e = 169.36K$ and $P_{ao} = 3.73 \times 10^{-10}\ W$ (B) $T_e = 170.8K$ and $P_{ao} = 4.56 \times 10^{-10}\ W$

(C) $T_e = 182.5K$ and $P_{ao} = 3.85 \times 10^{-10}\ W$ (D) $T_e = 160.62K$ and $P_{ao} = 4.6 \times 10^{-10}\ W$

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Q22. GATE ECE 2016 Set 1

Two lossless X-band horn antennas are separated by a distance of 200λ. The amplitude reflection coefficients at the terminals of the transmitting and receiving antennas are 0.15 and 0.18, respectively. The maximum directivities of the transmitting and receiving antennas (over the isotropic antenna) are $18\ dB$ and $22\ dB$, respectively. Assuming that the input power in the lossless transmission line connected to the antenna is $2\ W$, and that the antennas are perfectly aligned and polarization matched, the power (in mW) delivered to the load at the receiver is _____ .

Open solution

Q23. GATE ECE 2016 Set 1

The far-zone power density radiated by a helical antenna is approximated as:

$$\overrightarrow{W}_{rad} = \overrightarrow{W}\,average \approx \hat{a}_r C_0 \frac{1}{r^2}\cos^4\theta$$

The radiated power density is symmetrical with respect to ϕ and exists only in the upper hemisphere: $0 \le \theta \le \frac{\pi}{2};\quad 0 \le \theta \le 2\pi;$

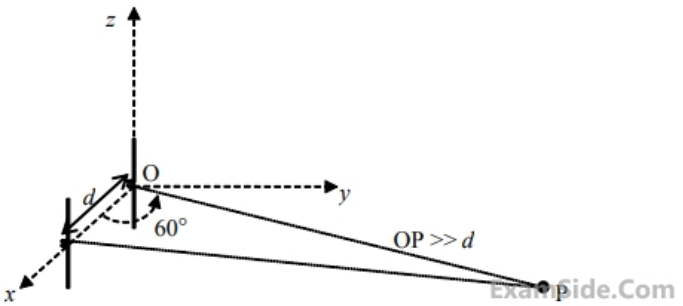
C_0 is a constant. The power radiated by the antenna (in watts) and the maximum directivity of the antenna, respectively, are

(A) $1.5C_0,\ 10dB$ (B) $1.256C_0,\ 10dB$ (C) $1.256C_0,\ 12dB$ (D) $1.5C_0,\ 12dB$

Open solution

Q24. GATE ECE 2015 Set 2

Two half-wave dipole antennas placed as shown in the figure are excited with sinusoidally varying currents of frequency 3 MHz and phase shift of $\pi/2$ between them (the element at the origin leads in phase). If the maximum radiated E-field at the point P in the x-y plane occurs at an azimuthal angle 60°, the distance d (in meters) between the antennas is _____ .



Open solution

Q25. GATE ECE 2008

At 20 GHz, the gain of a parabolic dish antenna of 1 meter diameter and 70% efficiency is

(A) 15 dB (B) 25 dB (C) 35 dB (D) 45 dB

Open solution

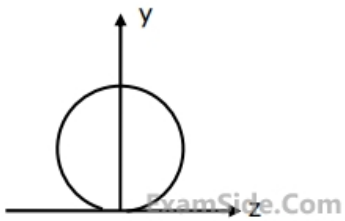
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Q26. GATE ECE 2007

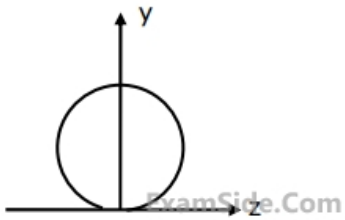
A $\lambda/2$ dipole is kept horizontally at a height of $\lambda_0/2$ above a perfectly conducting infinite ground plane. The radiation pattern in the plane of the dipole ($\vec{E}$ plane) looks approximately as

Open solution

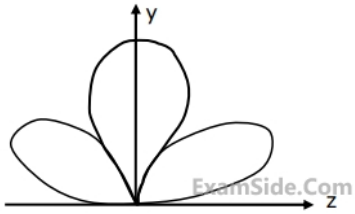
(A)



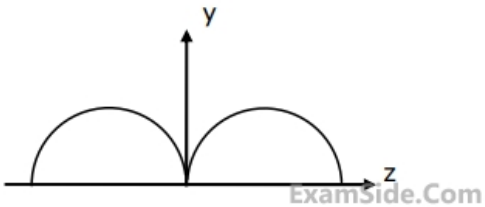
(B)



(C)



(D)



Q27. GATE ECE 2006

A mast antenna consisting of a 50 meter long vertical conductor operates over a perfectly conducting ground plane. It is base-fed at frequency of 600 kHz. The radiation resistance of the antenna in ohms is

Open solution

(A) $\frac{2\ \pi^2}{5}$ (B) $\frac{\pi^2}{5}$ (C) $\frac{4\ \pi^2}{5}$ (D) $20\ \pi^2$

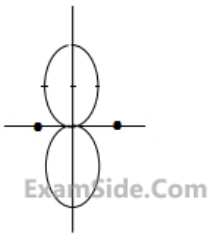
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Q28. GATE ECE 2005

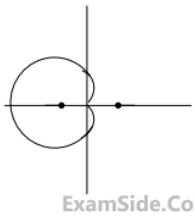
Two identical and parallel dipole antennas are kept apart by a distance $\lambda/4$ of in the H-plane. They are fed with equal currents but the right most antenna has a phase shift of +90°. The radiation pattern is given as

Open solution

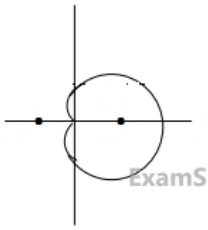
(A)



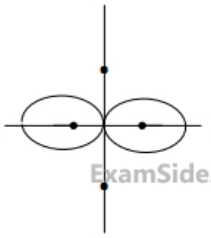
(B)



(C)



(D)



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Q29. GATE ECE 2003

Two identical antennas are placed in the $\theta = \pi/2$ plane as shown in figure. The elements have equal amplitude excitation with 180° polarity difference, operating at wavelength λ. The correct value of the magnitude of the far zone resultant electric field strength normalized with that of a single element, both computed for $\phi = 0$ is

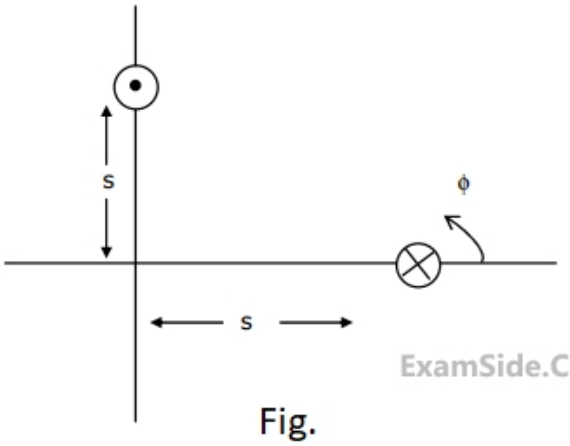


Fig.

(A) $2\cos\left(\frac{2\pi s}{\lambda}\right)$ (B) $2\sin\left(\frac{2\pi s}{\lambda}\right)$

(C) $2\cos\left(\frac{\pi s}{\lambda}\right)$ (D) $2\sin\left(\frac{\pi s}{\lambda}\right)$

Open solution

Q30. GATE ECE 2002

A person with a receiver is 5 Km away from the transmitter. What is the distance that this person must move further to detect a 3-dB decrease in signal strength?

(A) 942 m (B) 2070 m (C) 4978 m (D) 5320 m

Open solution

Q31. GATE ECE 2001

A medium wave radio transmitter operating at a wavelength of 492 m has a tower antenna of height 124m. What is the radiation resistance of the antenna?

(A) 25 Ω (B) 36.5 Ω (C) 50 Ω (D) 73 Ω

Open solution

Q32. GATE ECE 2001

In a uniform linear array, four isotropic radiating elements are spaced $\lambda/4$ apart. The progressive phase shift between the elements required for forming the main beam at 60° off the end-fire is

(A) $-\pi\ rad$ (B) $-\pi/2\ rad$ (C) $-\pi/4\ rad$ (D) $-\pi/8\ rad$

Open solution

Q33. GATE ECE 2000

For an 8 feet (2.4 m) parabolic disk antenna operating at 4 GHz, the minimum distance required for far field measurement is closest to

(A) 7.5 cm (B) 15 cm (C) 15 m (D) 150 m

Open solution

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Q34. GATE ECE 2000

The half-power beam widths (HPBW) of an antenna in the two orthogonal planes are 100° and 60° respectively. The directivity of the antenna is approximately equal to

(A) 2 db (B) 5 db (C) 8 db (D) 12 db

Open solution

Q35. GATE ECE 1999

A transmitting antenna radiates 251 W isotropically. A receiving antenna, located 100m away from the transmitting antenna, has an effective aperture of 500 cm^2. The total power received by the antenna is

(A) 10 nW (B) 1 μ W (C) 20 μ W (D) 100 μ W

Open solution

Q36. GATE ECE 1997

A parabolic dish antenna has a conical beam 2^0 wide. The directivity of the antenna is approximately

(A) 20 dB (B) 30 dB (C) 40 dB (D) 50 dB

Open solution

Q37. GATE ECE 1996

A transverse electromagnetic wave with circular polarization is received by a dipole antenna. Due to polarization mismatch, the power transfer efficiency from the wave to the antenna is reduced to about

(A) 50% (B) 35.3% (C) 25% (D) 05

Open solution

Q38. GATE ECE 1996

A 1 km long microwave link uses two antennas each having 30dB gain. If the power transmitted by one antenna is 1 W at 3 GHz, the power received by the other antenna is approximately

(A) 98.6 μ W (B) 76.8 μ W (C) 63.4 μ W (D) 55.2 μ W

Open solution

Q39. GATE ECE 1992

Two dissimilar antennas having their maximum directivities equal

(A) must have their beamwidths also equal

(B) cannot have their beamwidths equal because they are dissimilar antennas

(C) may not necessarily have their maximum power gains equal

(D) must have their effective aperture areas (capture areas) also equal.

Open solution

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Q40. GATE ECE 1992

The beam width-between-first-null of a uniform linear array of N equally-spaced (element spacing = d) equally-excited antennas is determined by

(A) N alone and not by d
(B) d alone and not by N
(C) the ratio (N/d)
(D) the product (Nd)

Open solution

Q41. GATE ECE 1991

In a board side array of 20 isotropic radiators, equally spaced at a distance of $\lambda/2$, the veam width between first nulls is

(A) 51.3 degrees
(B) 11.46 degrees
(C) 22.9 degrees
(D) 102.6 degrees

Open solution

Q42. GATE ECE 1990

Two isotropic antennas are separated by a distance of two wavelengths. If both the antennas are fed with currents of equal phase and magnitude, the number of lobes in the radiation pattern in the horizontal plane are

(A) 2
(B) 4
(C) 6
(D) 8

Open solution

Q43. GATE ECE 1989

In the case of ground wave propagation, at far-off distances from the transmitting tower antenna, the Poynting vector is

(A) in the truly horizontal direction
(B) essentially horizontal but for a small downward component
(C) essentially horizontal but for a small upward component
(D) directed upward at an angle determined by the "effective height" of the transmitting tower antenna.

Open solution

Q44. GATE ECE 1988

The electric field E and the magnetic field H of a short dipole antenna satisfy the condition

(A) the r component of E is equal to zero
(B) both r and θ components of H are equal to zero
(C) the θ component of E dominates the r component in the far-field region
(D) the θ and ϕ components of H are of the same order of magnitude in the near-field region

Open solution

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Marks 4 1 questions

Q45. GATE ECE 1987

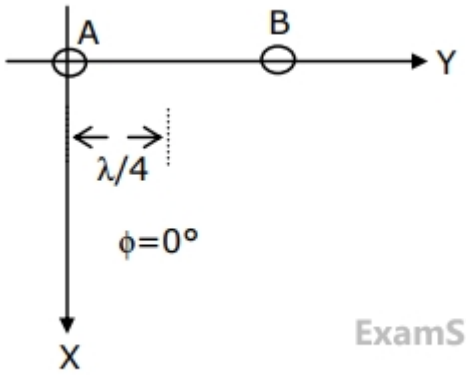
A thin dipole antenna is $(\lambda/15)$ long, where λ is the voperating wavelength. If the dipole has a loss resistance of 1.5 ohm determine its
(i) radiation resistance and
(ii) terminal resistance

Open solution

Marks 5 7 questions

Q46. GATE ECE 2002

Consider a linear array of two half - wave dipoles A and B as shown in Fig. The dipoles are $\frac{\lambda}{4}$ apart and are excited in such a way that the current on element B lags that on element A by 90^0 in phase
(a) Obtain the expression for the radiation pattern for E in the XY plane, i.e., $(\theta = 90^0)$.
(b) Sketch the radiation pattern obtained in (a).



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Open solution

Q47. GATE ECE 1999

The average power of an omni-directional antenna varies as the magnitude of cos(θ) where θ is the azimuthal angle. Calculate the maximum Directive Gain of the antenna and the angles at which it occurs.

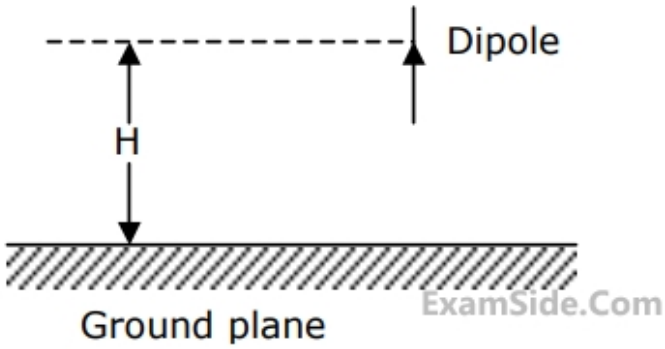
Open solution

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Q48. GATE ECE 1997

Open solution

A dipole antenna has a sinθ radiation pattern where the angle θ is measured from the axis of the dipole. The dipole is vertically located above an ideal ground plane (Fig.). What should be the height of the dipole H in terms of wave length so as to get a null in the radiation pattern at an angle of 45^0 from the ground plane? Find the direction of maximum radiation also.



Q49. GATE ECE 1996

Open solution

Two isotropic antennas A and B from an array as shown in Fig. The currents fed to the two antennas are $\mathrm{I}_0 \angle 0$ and $\mathrm{I}_0 \angle \alpha$ respectively. What should be the value of α so that the radiation pattern has a null at $\theta = 30^\circ$. Find the direction of the maximum radiation for that value of α and draw the radiation pattern. (λ is the wavelength of operation)

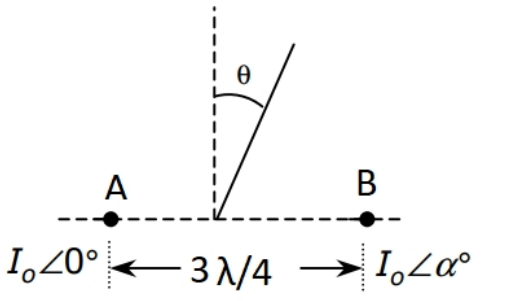


Fig.

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Q50. GATE ECE 1995

Open solution

Two dipoles are so fed and oriented in free space that they produce the following electromagnetic waves:

$$E_x = 10\, e^{j(\omega t - z\pi/3)}\ volts/metre$$

$$E_y = j\ 10\, e^{j(\omega t - z\pi/3)}\ volts/metre$$

(a) Write down the expression for the corresponding magnetic field strength vector.
(b) Calculate the frequency of the wave.
(C) Give the complete description of the polarization of the wave.

Q51. GATE ECE 1994

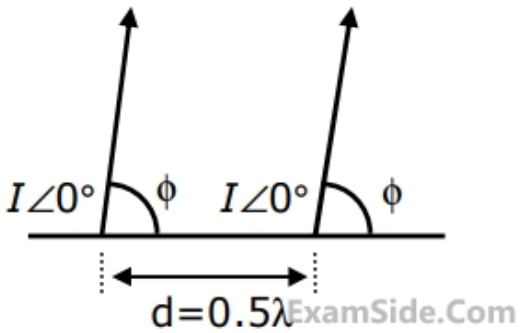
Open solution

Two spacecrafts are separated by 3000 km. Each has a paraboloidal reflector antenna of 0.85 m diameter operating at a frequency of 2 GHz with an aperture efficiency of 64%. If the space crafts A's receiver requires 1pW for a 20 dB signal-to-noise ratio, what transmitter power is required on the spacecraft B to achieve this signal-to-noise ratio?

Q52. GATE ECE 1993

Open solution

Consider an array of two non-directional radiators with spacing $d = 0.5\ \lambda$. Determine the directions of maximum radiation when the radiators are excited as shown in Fig.. Calculate the phase shift required for turning the direction of the maximum radiation by 90° keeping the separation d unchanged.



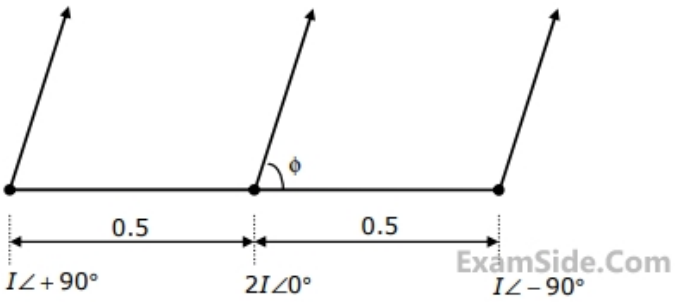
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Marks 8 1 questions

Q53. GATE ECE 1992

Elements of a linear array of three equally spaced (element spacing 0.5λ) vertical mast radiators are excited as given in Fig. For the horizontal plane radiation pattern of the array determine the direction of the major lobe (main lobe or principal lobe) and calculate its half-power-beam-width in degrees.

Open solution



Marks 10 2 questions

Q54. GATE ECE 1991

In the radiation pattern of a 3-element array of isotropic radiators equally spaced at distances of $\lambda/4$ it is required to place a null at an angle of 33.56 degrees off the end-fire direction. Calculate the progressive phase shifts to be applied to the elements. Also calculate the angle at which the main beam is placed for this phase distribution.

Open solution

Q55. GATE ECE 1989

Two vertically oriented half wave dipoles are spaced 1.5 wavelengths apart to from an array. Calculate the half power beam widths of the major lobes of the array, in the horizontal plane for the following two cases when the dipoles are fed with

(a) equal and in - phase currents (Broadside array)

(b) equal currents but with a phase difference of 540^0 between the two currents (End fire array)

Open solution

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Answer Key

Marks 1

1. **B**	2. **0.24 to 0.25**	3. **B**	4. **47.7**	5. **B**
6. **B**	7. **C**	8. **A**	9. **C**	10. **A**
11. **B**	12. **C**	13. **D**	14. **C**	15. **A**
16. **C**	17. **C**	18. **A**		

Marks 2

19. **D**	20. **A**	21. **A**	22. **2.9**	23. **B**
24. **50**	25. **D**	26. **C**	27. **A**	28. **B**
29. **D**	30. **B**	31. **B**	32. **C**	33. **D**
34. **C**	35. **D**	36. **C**	37. **D**	38. **C**
39. **C**	40. **D**	41. **B**	42. **D**	43. **C**
44. **B, C**				

Marks 4

45. **(i)** $20\pi^2\left(\frac{\ell}{\lambda}\right)^2$ **(ii)** $1.5 + 20\pi^2\left(\frac{\ell}{\lambda}\right)^2$

Marks 5

46. **(a)** $2\cos\left(\frac{\Psi}{2}\right)$, $\Psi = \frac{\pi}{2}\sin\phi - \frac{\sigma}{2}$	47. **Maximum Directive Gain** $= 4$ **and angles are** 0° **and** 180°	48. $H = \frac{\lambda}{2\sqrt{2}}$ **Direction of maximum radiation** $H = \frac{\lambda}{2\sqrt{2}}$	49. $\alpha = \frac{\pi}{4}$ **Direction of maximum radiation** $= \pm\ 99.6^\circ$	50. **(b) 50 MHz (c) This is left circular polarization**
51. **Transmitter power at B = 1.535 W.**	52. **Directions of maximum radiation are** $= \pm 90^\circ$ **, Phase shift** $= 180^\circ$			

Marks 8

53. **Direction of major lobe** $= \pm 60^\circ$ **Half-power-beam-width** $= 52^\circ$

Marks 10

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54. **Progressive phase sifts $= 45^\circ$, Main beam placed at angle $= 120^\circ$**	55. **(a)19.2^0 (b) 67.2^0**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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